

Documents Test Driven refactoring of the kata

1. Run default test

```

===== test session starts =====
platform linux -- Python 3.8.3, pytest-5.4.3, py-1.9.0, pluggy-0.13.1
rootdir: /home/xavyjs/projects/GildedRose-Refactoring-Kata/python
plugins: approvaltests-0.2.4, approvaltests-14.7.1
collected 1 item

tests/test_gilded_rose.py F                                     [100%]

===== FAILURES =====
_____ GildedRoseTest.test_foo _____

self = <tests.test_gilded_rose.GildedRoseTest testMethod=test_foo>

    def test_foo(self):
        items = [Item("foo", 0, 0)]
        gilded_rose = GildedRose(items)
        gilded_rose.update_quality()
> self.assertEqual("fixme", items[0].name)
E       AssertionError: 'fixme' != 'foo'
E       - fixme
E       + foo

tests/test_gilded_rose.py:12: AssertionError
===== short test summary info =====
FAILED tests/test_gilded_rose.py::GildedRoseTest::test_foo - AssertionError: 'fixme' != 'foo'
===== 1 failed in 0.10s =====

```

Fix: Correct method name 'fixme' to 'foo'

1. Check Item Name is constant always

```

(GildedRose-Refactoring-Kata) (base) xavyjs@den:~/projects/GildedRose-Refactoring-Kata/python$ pytest
tests/test_gilded_rose.py
===== test session starts =====
platform linux -- Python 3.8.3, pytest-5.4.3, py-1.9.0, pluggy-0.13.1
rootdir: /home/xavyjs/projects/GildedRose-Refactoring-Kata/python
plugins: approvaltests-0.2.4, approvaltests-14.7.1
collected 1 item

tests/test_gilded_rose.py .                                     [100%]

===== 1 passed in 0.04s =====

```

2. Add test cases for base requirements

2. Check Item SellIn

3. Check Item Quality

```

tests/test_gilded_rose.py
===== test session starts =====
platform linux -- Python 3.8.3, pytest-5.4.3, py-1.9.0, pluggy-0.13.1
rootdir: /home/xavyjs/projects/GildedRose-Refactoring-Kata/python
plugins: approvaltests-0.2.4, approvaltests-14.7.1
collected 3 items

tests/test_gilded_rose.py ...                                   [100%]

===== 3 passed in 0.03s =====

```

3. Add test cases for Finer Constraints

4. Check Item Quality is never Negative**5. Check Item Quality is never more than 50****6. Check Item Quality degrade after SellIn**

```
===== test session starts =====
platform linux -- Python 3.8.3, pytest-5.4.3, py-1.9.0, pluggy-0.13.1
rootdir: /home/xavyjs/projects/GildedRose-Refactoring-Kata/python
plugins: approvaltests-0.2.4, approvaltests-14.7.1
collected 6 items

tests/test_gilded_rose.py ..... [100%]

===== 6 passed in 0.03s =====
```

4. Major refactor of glided_rose and test_glided_rose**1. Refactor GlidedRose Class to python function as the Class only has behavior and data is not native (instantiated by a different class i.e. Item)****2. Move Item class to a separate .py file for easier maintenance****3. Adding Type Hints for better readability on 1 and 2****4. Refactor test_glided_rose to adapt to changes from 1 to 3**

```
===== test session starts =====
platform linux -- Python 3.8.3, pytest-5.4.3, py-1.9.0, pluggy-0.13.1
rootdir: /home/xavyjs/projects/GildedRose-Refactoring-Kata/python
plugins: approvaltests-0.2.4, approvaltests-14.7.1
collected 6 items

tests/test_gilded_rose.py ..... [100%]
```

5. Adding Tests for straight forward cases**1. Aged Brie****2. Sulfuras**

```
===== test session starts =====
platform linux -- Python 3.8.3, pytest-5.4.3, py-1.9.0, pluggy-0.13.1
rootdir: /home/xavyjs/projects/GildedRose-Refactoring-Kata/python
plugins: approvaltests-0.2.4, approvaltests-14.7.1
collected 9 items

tests/test_aged_brie.py . [ 11%]
tests/test_sulfuras.py .. [ 33%]
tests/test_gilded_rose.py ..... [100%]

===== 9 passed in 0.02s =====
```

6. Adding Tests for backstage**1. Increase in Quality w.r.t SellIn****2. Quality +2 when days leq 10****3. Quality +3 when days leq 5****4. Quality drops to 0 post concert**

```
===== test session starts =====
platform linux -- Python 3.8.3, pytest-5.4.3, py-1.9.0, pluggy-0.13.1
rootdir: /home/xavyjs/projects/GildedRose-Refactoring-Kata/python
plugins: approvaltests-0.2.4, approvaltests-14.7.1
collected 4 items

tests/test_backstage.py .... [100%]

===== 4 passed in 0.01s =====
```

7. Refactoring gilded_rose

1. Separating loops and iter instance

```
===== test session starts =====
platform linux -- Python 3.8.3, pytest-5.4.3, py-1.9.0, pluggy-0.13.1
rootdir: /home/xavyjs/projects/GildedRose-Refactoring-Kata/python
plugins: approvaltests-0.2.4, approvaltests-14.7.1
collected 13 items

tests/test_aged_brie.py . [ 7%]
tests/test_backstage.py .... [ 38%]
tests/test_gilded_rose.py ..... [ 84%]
tests/test_sulfuras.py .. [100%]

===== 13 passed in 0.01s =====
```

2. Re-ordering setters for sell in and quality

```
===== test session starts =====
platform linux -- Python 3.8.3, pytest-5.4.3, py-1.9.0, pluggy-0.13.1
rootdir: /home/xavyjs/projects/GildedRose-Refactoring-Kata/python
plugins: approvaltests-0.2.4, approvaltests-14.7.1
collected 13 items

tests/test_aged_brie.py . [ 7%]
tests/test_backstage.py .... [ 38%]
tests/test_gilded_rose.py ..... [ 84%]
tests/test_sulfuras.py .. [100%]

===== 13 passed in 0.01s =====
```

3. Replacing Item Strings by Constants to avoid frequent typos

```
===== test session starts =====
platform linux -- Python 3.8.3, pytest-5.4.3, py-1.9.0, pluggy-0.13.1
rootdir: /home/xavyjs/projects/GildedRose-Refactoring-Kata/python
plugins: approvaltests-0.2.4, approvaltests-14.7.1
collected 13 items

tests/test_aged_brie.py . [ 7%]
tests/test_backstage.py .... [ 38%]
tests/test_gilded_rose.py ..... [ 84%]
tests/test_sulfuras.py .. [100%]

===== 13 passed in 0.06s =====
```

4. Refactor nested item quality checks with functions

```
===== test session starts =====
platform linux -- Python 3.8.3, pytest-5.4.3, py-1.9.0, pluggy-0.13.1
rootdir: /home/xavyjs/projects/GildedRose-Refactoring-Kata/python
plugins: approvaltests-0.2.4, approvaltests-14.7.1
collected 13 items

tests/test_aged_brie.py . [ 7%]
tests/test_backstage.py .... [ 38%]
tests/test_gilded_rose.py ..... [ 84%]
tests/test_sulfuras.py .. [100%]

===== 13 passed in 0.06s =====
```

5. Increase readability of conditionals with inversion of condition

```
===== test session starts =====
platform linux -- Python 3.8.3, pytest-5.4.3, py-1.9.0, pluggy-0.13.1
rootdir: /home/xavyjs/projects/GildedRose-Refactoring-Kata/python
plugins: approvaltests-0.2.4, approvaltests-14.7.1
collected 13 items

tests/test_aged_brie.py .                      [ 7%]
tests/test_backstage.py ....                  [ 38%]
tests/test_gilded_rose.py .....              [ 84%]
tests/test_sulfuras.py ..                     [100%]

===== 13 passed in 0.06s =====
```

6. Simplify conditionals by collapsing logic

```
===== test session starts =====
platform linux -- Python 3.8.3, pytest-5.4.3, py-1.9.0, pluggy-0.13.1
rootdir: /home/xavyjs/projects/GildedRose-Refactoring-Kata/python
plugins: approvaltests-0.2.4, approvaltests-14.7.1
collected 13 items

tests/test_aged_brie.py .                      [ 7%]
tests/test_backstage.py ....                  [ 38%]
tests/test_gilded_rose.py .....              [ 84%]
tests/test_sulfuras.py ..                     [100%]

===== 13 passed in 0.06s =====
```

7. Abstractions to simplify and separate concerns to ease the introduction of new item types

1. Introduce ItemUpdater abstraction via Protocol
2. Define DefaultItemUpdater behaviors
3. For every item type inherit DefaultItemUpdater and override behaviors according to the SRS

```
===== test session starts =====
platform linux -- Python 3.8.3, pytest-5.4.3, py-1.9.0, pluggy-0.13.1
rootdir: /home/xavyjs/projects/GildedRose-Refactoring-Kata/python
plugins: approvaltests-0.2.4, approvaltests-14.7.1
collected 13 items

tests/test_aged_brie.py .                      [ 7%]
tests/test_backstage.py ....                  [ 38%]
tests/test_gilded_rose.py .....              [ 84%]
tests/test_sulfuras.py ..                     [100%]

===== 13 passed in 0.06s =====
```

8. Test for conjured

Add test case for conjured

```
===== test session starts =====
platform linux -- Python 3.8.3, pytest-5.4.3, py-1.9.0, pluggy-0.13.1
rootdir: /home/xavyjs/projects/GildedRose-Refactoring-Kata/python
plugins: approvaltests-0.2.4, approvaltests-14.7.1
collected 14 items

tests/test_aged_brie.py . [ 7%]
tests/test_backstage.py .... [ 35%]
tests/test_conjured.py F [ 42%]
tests/test_gilded_rose.py ..... [ 85%]
tests/test_sulfuras.py .. [100%]

===== FAILURES =====
_____ test_conjured_degrades_twice_as_fast _____

    def test_conjured_degrades_twice_as_fast():
        item = Item(CONJURED, 0, 10)
        update_quality([item])
        assert 6 == item.quality
>
E       assert 6 == 8
E       + where 8 = Conjured Mana Cake, -1, 8.quality

tests/test_conjured.py:12: AssertionError
===== short test summary info =====
FAILED tests/test_conjured.py::test_conjured_degrades_twice_as_fast - assert 6 == 8
===== 1 failed, 13 passed in 0.13s =====
```

Test Failed

9. Feat: Add support for conjured item type

Introduce necessary constants and ConjuredUpdater to decrease twice as faster

```
===== test session starts =====
platform linux -- Python 3.8.3, pytest-5.4.3, py-1.9.0, pluggy-0.13.1
rootdir: /home/xavyjs/projects/GildedRose-Refactoring-Kata/python
plugins: approvaltests-0.2.4, approvaltests-14.7.1
collected 14 items

tests/test_aged_brie.py . [ 7%]
tests/test_backstage.py .... [ 35%]
tests/test_conjured.py . [ 42%]
tests/test_gilded_rose.py ..... [ 85%]
tests/test_sulfuras.py .. [100%]

===== 14 passed in 0.07s =====
```

Test Passes!

10. Refactor(gilded_rose): Decompose into independent modules for easier maintenance

1. config.py - contains item string definitions, MIN and MAX values for QUALITY
2. quality_operation - moved the functions for quality increase and decrease into a separate module
3. item_updater.py - contains ItemUpdater abstract, DefaultItemUpdater definition, Overrided behaviors of DefaultItemUpdater for every item type
4. gilded_rose.py - glues 1 to 3

```
===== test session starts =====
platform linux -- Python 3.8.3, pytest-5.4.3, py-1.9.0, pluggy-0.13.1
rootdir: /home/xavyjs/projects/GildedRose-Refactoring-Kata/python
plugins: approvaltests-0.2.4, approvaltests-14.7.1
collected 14 items

tests/test_aged_brie.py .                      [  7%]
tests/test_backstage.py ....                  [ 35%]
tests/test_conjured.py .                      [ 42%]
tests/test_gilded_rose.py .....              [ 85%]
tests/test_sulfuras.py ..                    [100%]

===== 14 passed in 0.06s =====
```

Test Passes post refactoring into seperate modules!

TO INTRODUCE NEW ITEM TYPES (like conjured)

1. Add the item types string in config.py
2. In item_updater.py, Add the behavior for the new item type by overriding the default behaviors from DefaultItemUpdater
3. Update the new item type in gilded_rose.py